Danny Yu

Computer Engineering Graduate · Internal Platforms for STEM Education

Vancouver, BC · danny_yu_2@sfu.ca · yannydu.github.io · github.com/yannydu

Summary

Computer Engineering graduate (SFU, 2025) who started and leads the internal web platforms at SFU Science Alive, built by directing AI coding agents: I set what each change must do, check each release against what staff need, and refine from user feedback. Because the users are children, much of the work is privacy rules: real names are never stored. Earlier: two QA and test-automation co-ops.

Skills

Languages:	Python, JavaScript, HTML / CSS, Typst
Tools:	Git and GitHub (issues, pull requests), Linux, Claude Code
LLM workflows:	prompt engineering (Claude Code skills, agent instructions, hooks), reviewing LLM output against requirements
Product leadership:	requirements and priorities, privacy rules for minors, user support, documentation

Work Experience

SFU Science Alive (Faculty of Applied Sciences Outreach) — *Burnaby, BC*
School Programs and Instructional Lead — *Jun. 2021 – Present*

- Started and lead the program's internal web platforms, built with Claude Code: I set the requirements and priorities, and check each release against what staff need. Privacy rules for children: camper names become pseudonyms before anything is stored, and camper code runs apart from the main site.
- Designed the AI workflow behind the build: Claude Code skills that take a request from board item to merged change, hooks that block risky actions, and specialized agents for building, review, and QA. Analyzed 365 AI-agent runs to find what drove cost and turn count, then changed brief limits and model settings from the results.
- Launched **spark-gallery**, an online magazine of camper projects published each camp week: 509 projects from 471 campers through a full summer of camps.
- Introduced **campkit**, a shared curriculum platform that replaced my earlier document pipeline, holding 356 activities and 25 camps for a six-person staff team.
- Grew **hello-wattson** (hellowattson.ca) from an internal lesson site into a code-gated teaching platform for outside teachers that collects no student data: 22 programs, with an in-browser Arduino compiler and code playground.
- Keep refining the platforms from feedback: triage problems reported by writers and instructors into tracked issues, document them, and decide what ships next.
- Wrote SPARK, the program's workshop-design framework; teach coding camps (Python, web, Arduino, LEGO robotics).

Global Relay — *Vancouver, BC*
Software Development Engineer in Test (Co-op) — *Jan. 2023 – Aug. 2023*

- Built a Java JSON response builder for API validation and generated Java bindings from Thrift IDLs, integrating them with Wiremock containers for test stubs.
- Implemented a single source of truth for regression tests using the Trino REST API.

Incognito Software Systems Inc. — *Vancouver, BC*
QA Engineer Intern — *Jan. 2022 – Aug. 2022*

- Wrote automated GUI tests in Java and Selenium and set up a Docker + Jenkins continuous-testing environment.

Projects

ChessMate: A Smart Automated Chessboard — *Burnaby, BC*
SFU ENSC Capstone — *Jan. 2024 – Aug. 2024*

- Built the human-move detection module with a Raspberry Pi camera and OpenCV, comparing board states to identify the moved piece; drove an electromagnetic trolley in Python to execute the machine's moves.

Education

Simon Fraser University — *Burnaby, BC*
BASc in Computer Engineering — *Graduated 2025*